CIRCULAR QUEUE USING ARRAY - C PROGRAM

1. PROGRAM CODE

```
#include <stdio.h>
#define MAX 5
int queue[MAX];
int front = -1, rear = -1;
void insert()
{
    int num;
    if (front == (rear + 1) % MAX)
    {
        printf("Queue Overflow\n");
        return;
    }
    printf("Enter number to insert: ");
    scanf("%d", &num);
    if (front == -1)
        front = rear = 0;
    else
        rear = (rear + 1) % MAX;
    queue[rear] = num;
    printf("%d Inserted\n", num);
}
void delete()
{
    int num;
    if (front == -1)
    {
        printf("Queue Underflow\n");
        return;
    }
    num = queue[front];
    printf("%d deleted\n", num);
    if (front == rear)
        front = rear = -1;
    else
        front = (front + 1) % MAX;
}
void display()
{
    int i;
    if (front == -1)
    {
        printf("Queue is Empty\n");
        return;
    }
    printf("Queue: ");
    for (i = front; ; i = (i + 1) % MAX)
    {
        printf("%d ", queue[i]);
        if (i == rear)
            break;
    }
    printf("\n");
}
int main()
{
    int choice;
    while (1)
    {
        printf("\n1. Insert\n");
        printf("2. Delete\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
                insert();
                break;
            case 2:
                delete();
                break;
            case 3:
                display();
                break;
            case 4:
                return 0;
            default:
                printf("Invalid Choice\n\n");
        }
    }
    return 0;
}
```

2. ACTUAL PROGRAM OUTPUT

Representative full run (captured stdout)

```
1. Insert
2. Delete
3. Display
4. Exit
Enter choice: Enter number to insert: 10 Inserted

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: Enter number to insert: 20 Inserted

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: Queue: 10 20

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: 10 deleted

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: Enter number to insert: 30 Inserted

1. Insert
2. Delete
3. Display
4. Exit
Enter choice: Queue: 20 30

1. Insert
2. Delete
3. Display
4. Exit
Enter choice:
```